

End Term Solution July-2023

PG & P-202

Sol. 1-(a) Let there are three Urns say U_1, U_2 and U_3 .

6R 4B	4R 6B	5R 5B
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Let $E_1 = \text{Urn } U_1 \text{ is chosen}$ $U_1 \quad U_2 \quad U_3$

$E_2 = \text{Urn } U_2 \text{ is chosen}$

$E_3 = \text{Urn } U_3 \text{ is chosen.}$

A : Ball drawn is a red ball.

Since there are three urns and one of them is chosen at random, therefore

$$P(E_1) = P(E_2) = P(E_3) = \frac{1}{3}$$

$$\text{Also } P(A|E_1) = \frac{6}{10}, \quad P(A|E_2) = \frac{4}{10},$$

$$P(A|E_3) = \frac{5}{10}$$

Required probability

$$\begin{aligned}
 P(E_1|A) &= \frac{P(E_1) \cdot P(A|E_1)}{P(E_1) \cdot P(A|E_1) + P(E_2) \cdot P(A|E_2) + P(E_3) \cdot P(A|E_3)} \\
 &= \frac{\frac{1}{3} \times \frac{6}{10}}{\frac{1}{3} \left[\frac{6}{10} + \frac{4}{10} + \frac{5}{10} \right]} = \frac{\frac{6}{10}}{\frac{15}{10}} = \frac{2}{5}
 \end{aligned}$$

1.b. Let the manufactured produces x_1 unit of product M_1 and x_2 unit of model M_2

Formulation:-

	M_1	M_2	Availability
Grinding	4	2	$2 \times 40 = 80$
polishing	2	5	$3 \times 40 = 120$
Profit	3	4	

Formulation:- Max $Z = 3x_1 + 4x_2$

Subject to $4x_1 + 2x_2 \leq 80$
 $2x_1 + 5x_2 \leq 120$

Restriction $x_1, x_2 \geq 0$.

(c) Determine the value of c that makes the function $f(x, y) = c(x+y)$, a joint PMF over $x = 1, 2, 3$, $y = 1, 2, 3$. Find $E[X]$ and $E[Y]$

$x \backslash y$	1	2	3	$f(x)$
1	$2c$	$3c$	$4c$	$9c$
2	$3c$	$4c$	$5c$	$12c$
3	$4c$	$5c$	$6c$	$15c$
$f(y)$	$9c$	$12c$	$15c$	$36c$

Sample Space is $\left\{ \begin{array}{l} (1,1) \ (1,2) \ (1,3) \\ (2,1) \ (2,2) \ (2,3) \\ (3,1) \ (3,2) \ (3,3) \end{array} \right\}$

Since $f(x, y) = c(x+y)$ is PMF if

$$\sum f(x, y) = 1 \Rightarrow 36c = 1 \Rightarrow c = \frac{1}{36}$$

$$E[X] = 1 \times 9c + 2 \times 12c + 3 \times 15c$$

$$= 9c + 24c + 45c = 78c = \frac{78}{36}$$

$$E[Y] = 9c + 24c + 15c = \frac{78}{36} \Rightarrow c = \frac{1}{36}$$

1-(d) Give $n=18$, $\sum x = 12$, $\sum y = 18$, $\sum x^2 = 60$
 $\sum y^2 = 96$, $\sum xy = 48$

Regression coefficient y on x

$$b_{yx} = \frac{n \sum xy - \sum x \sum y}{n \sum x^2 - (\sum x)^2} = \frac{18 \times 48 - 12 \times 18}{18 \times 60 - (12)^2}$$

$$= \frac{864 - 216}{1080 - 144} = \frac{648}{936} = 0.69$$

Regression y on x is

$$y - \bar{y} = b_{yx}(x - \bar{x}) \Rightarrow y - 1 = 0.69(x - 0.67)$$

$$\therefore \bar{x} = \frac{\sum x}{n}, \bar{y} = \frac{\sum y}{n} = 1 \quad \left| \quad \begin{aligned} y - 1 &= 0.69x - 0.4623 \\ 0.69x - y &= -0.5377 = 0 \end{aligned} \right.$$

Regression x on y , regression coefficient

$$b_{xy} = \frac{n \sum xy - \sum x \sum y}{n \sum y^2 - (\sum y)^2} = \frac{18 \times 48 - 12 \times 18}{18 \times 96 - (18)^2}$$

$$= \frac{648}{1728 - 324} = \frac{648}{1404} = 0.46$$

Regression x on y is

$$x - \bar{x} = b_{xy}(y - \bar{y}) \text{ or } x - 0.67 = 0.46(y - 1)$$

$$\text{or } x - 0.67 = 0.46y - 0.46 \quad \text{or } x + 0.46y = 0.67$$

$$\boxed{x - 0.46y = 0.23} \quad -0.46$$

$$\frac{1-e}{M_X(t)} = \sum_{n=1}^{\infty} e^{tn} \cdot f(n) = \sum_{n=1}^{\infty} e^{tn} \frac{1}{2^n}$$

put $z = e^t$, $\sum_{n=1}^{\infty} \frac{z^n}{2^n} = \sum_{n=1}^{\infty} \left(\frac{z}{2}\right)^n$

$$= \left(\frac{z}{2}\right) + \left(\frac{z}{2}\right)^2 + \left(\frac{z}{2}\right)^3 + \dots \infty$$

$$= \left[\left(\frac{e^t}{2}\right) + \left(\frac{e^t}{2}\right)^2 + \left(\frac{e^t}{2}\right)^3 + \dots \right]$$

$$= \frac{e^t}{2} \left[1 + \frac{e^t}{2} + \left(\frac{e^t}{2}\right)^2 + \left(\frac{e^t}{2}\right)^3 + \dots \right]$$

$$= \frac{e^t}{2} \left[\left(1 - \frac{e^t}{2}\right)^{-1} \right] \quad \because (1-x)^{-1} = 1+x+\dots$$

$$= \frac{e^t}{2} \left[\left(\frac{2-e^t}{2}\right)^{-1} \right]$$

$$= \frac{e^t}{2} \left(\frac{2}{2-e^t} \right) = \frac{e^t}{2-e^t}$$

$$\therefore M_X(t) = \frac{e^t}{2-e^t} \quad \underline{\text{Ans.}}$$

2. (a) Poisson distribution as a limiting case of Binomial distribution.

$n \rightarrow \infty$, $p \rightarrow 0$ and $np = \lambda$ a finite quantity.

From Binomial distribution.

$$P(x) = {}^nC_x p^x q^{n-x}, \quad P(0) = {}^nC_0 p^0 q^{n-0} = q^n$$

$$\begin{aligned} \lim_{n \rightarrow \infty} P(0) &= \lim_{n \rightarrow \infty} q^n \\ &= \lim_{n \rightarrow \infty} (1-p)^n = \lim_{n \rightarrow \infty} \left(1 - \frac{\lambda}{n}\right)^n \\ &= e^{-\lambda} \quad \text{--- (1)} \end{aligned}$$

$$\text{Now, } \frac{P(x+1)}{P(x)} = \frac{{}^nC_{x+1} \cdot p^{x+1} q^{n-(x+1)}}{{}^nC_x \cdot p^x q^{n-x}}$$

$$= \frac{n-x}{x+1} \cdot \frac{p}{q}$$

$$\lim_{n \rightarrow \infty} \frac{P(x+1)}{P(x)} = \lim_{n \rightarrow \infty} \frac{n-x}{x+1} \cdot \frac{\lambda/n}{(1-\lambda/n)} \quad \because np = \lambda$$

$$= \lim_{n \rightarrow \infty} \frac{(1 - x/n) \cdot \lambda}{(1 - \lambda/n) \cdot (x+1)}$$

$$\therefore \lim_{n \rightarrow \infty} \frac{P(x+1)}{P(x)} = \frac{\lambda}{x+1}$$

--- (2)

$$\text{Now, } P(x) = \frac{P(x)}{P(x-1)} \cdot \frac{P(x-1)}{P(x-2)} \cdot \frac{P(x-2)}{P(x-3)} \dots \frac{P(1)}{P(0)} \cdot P(0)$$

Taking the limit when $n \rightarrow \infty$, using result (1) and (2), we get

$$\begin{aligned} \lim_{n \rightarrow \infty} P(x) &= \frac{\lambda}{x} \cdot \frac{\lambda}{x-1} \cdot \frac{\lambda}{x-2} \dots \frac{\lambda}{1} \cdot e^{-\lambda} \\ &= \frac{\lambda^x \cdot e^{-\lambda}}{x!} \end{aligned}$$

Hence the probability of x success is given by

$$P(x) = \frac{e^{-\lambda} \lambda^x}{x!}, \quad x = 0, 1, 2, 3, \dots$$

This is required results.

$$P(x=2) = 9 P(x=4) + 90 P(x=6)$$

$$\frac{e^{-\lambda} \lambda^2}{2!} = 9 \cdot \frac{e^{-\lambda} \lambda^4}{4!} + 90 \frac{e^{-\lambda} \lambda^6}{6!}$$

$$\text{or } \frac{1}{2!} = \frac{9 \cdot \lambda^2}{4!} + \frac{90 \cdot \lambda^4}{6!} \Rightarrow \lambda^4 + 3\lambda^2 - 4 = 0$$

$$(\lambda^2 + 4)(\lambda^2 - 1) = 0 \Rightarrow \lambda = 1. \quad \because \lambda > 0$$

For poisson distribution mean = variance = λ
 $= 1$

Ans 2(b):

Let $X \sim N(\mu, \sigma^2)$, then we are given

$$P(X < 45) = 0.31, \quad P(X > 64) = 0.08$$

$$\Rightarrow P[45 < X < \mu] = 0.50 - 0.31 = 0.19$$

$$P[\mu < X < 64] = 0.50 - 0.08 = 0.42$$

$$\Rightarrow P\left[\frac{45 - \mu}{\sigma} < \frac{X - \mu}{\sigma} < 0\right] = 0.19 \quad \text{and}$$

$$P\left[0 < \frac{X - \mu}{\sigma} < \frac{64 - \mu}{\sigma}\right] = 0.42$$

$$\Rightarrow P\left[0 < Z < \frac{\mu - 45}{\sigma}\right] = 0.19 \quad \text{and}$$

$$P\left[0 < Z < \frac{64 - \mu}{\sigma}\right] = 0.42$$

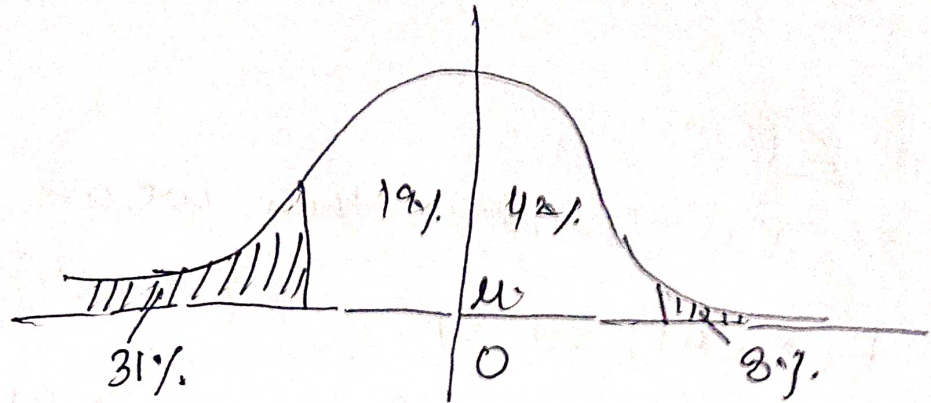
$$\therefore P[-Z < Z < 0] = P[0 < Z < Z]$$

From the table,

$$\frac{\mu - 45}{\sigma} = 0.496 \quad \text{and} \quad \frac{64 - \mu}{\sigma} = \underline{1.45}$$

$$\text{Adding, we get} \quad \frac{19}{\sigma} = 1.901 \Rightarrow \sigma = 10 (\text{approx})$$

$$\begin{aligned} \mu &= 45 + (0.496) \times 10 \\ &= 50 (\text{approx}) \end{aligned}$$



Let μ be the mean and σ be S.D

for $x=45$, $z = \frac{45-\mu}{\sigma}$

for $x=64$ $z = \frac{64-\mu}{\sigma}$

Area between 0 ~~to~~ z and $z = \frac{45-\mu}{\sigma} = 0.50 - 0.31 = 0.19$

From the table, for the area 0.19, $z = 0.496$

$$\frac{45-\mu}{\sigma} = -0.496$$

Area between $z=0$ and $z = \frac{64-\mu}{\sigma} = 0.50 - 0.08 = 0.42$

From the table, for area 0.42 $z = 1.408$

$$\frac{64-\mu}{\sigma} = 1.405$$

Solving, we get $\mu=50$ $\sigma=10$.

3(a) Given chance that a student will pass a statistics course $p = 0.85$

$$\text{mean} = np = 0.85 \times 120 = 102$$

$$q = 1 - p = 0.15$$

$$n = 120$$

~~$$P\left[\frac{X - \mu}{\sigma} \geq \frac{110 - 102}{\sigma = \sqrt{npq}}\right]$$~~

$$P\left[\frac{X - \mu}{\sigma} \geq \frac{110 - 102}{\sqrt{120 \times 0.85 \times 0.15}}\right]$$

$$= P\left[Z \geq \frac{8}{3.91152}\right] = P[Z \geq 2.045]$$

$$= P[0 \leq Z \leq 110]$$

$$= 0.5 - 2.045$$

3-(b) Since X follows uniform distribution.

$\Rightarrow$ PDF is given by

$$f(x) = \frac{1}{b-a}, \quad a < x < b$$

$$\text{Therefore } f(x) = \frac{1}{2-(-2)} \quad -2 < x < 2$$
$$= \frac{1}{4}, \quad -2 < x < 2$$

$$\text{li) } P[X < 0] = \int_{-2}^0 f(x) dx = \int_{-2}^0 \frac{1}{4} dx$$
$$= \frac{1}{4} [x]_{-2}^0$$
$$= \frac{1}{4} [0 - (-2)] = \frac{1}{2}$$

$$\text{lii) } P[|X-1| < \frac{1}{2}] = P[-\frac{1}{2} < X-1 < \frac{1}{2}]$$
$$= P[-\frac{1}{2}+1 < X < \frac{1}{2}+1]$$
$$= P[\frac{1}{2} < X < \frac{3}{2}]$$
$$= \int_{\frac{1}{2}}^{\frac{3}{2}} f(x) dx$$
$$= \int_{\frac{1}{2}}^{\frac{3}{2}} \frac{1}{4} dx = \frac{1}{4} [x]_{\frac{1}{2}}^{\frac{3}{2}}$$
$$= \frac{1}{4} [\frac{3}{2} - \frac{1}{2}] = \frac{1}{4}$$

4(a) $f(x, y) = e^{-x-y} \quad 0 < x, \quad 0 < y$

$$\begin{aligned}
 E[X] &= \int_0^{\infty} \int_0^{\infty} x f(x, y) \, dx \, dy \\
 &= \int_0^{\infty} \int_0^{\infty} x \cdot e^{-x-y} \, dx \, dy \\
 &= \int_0^{\infty} e^{-y} \, dy \int_0^{\infty} x e^{-x} \, dx \\
 &= \left[\frac{e^{-y}}{-1} \right]_0^{\infty} \left[\frac{x e^{-x}}{-1} - e^{-x} \right]_0^{\infty} \\
 &= -[0-1][0-\{0-1\}] = 1
 \end{aligned}$$

Similarly $E[Y] = 1$

$$\begin{aligned}
 E[X \cdot Y] &= \int_0^{\infty} \int_0^{\infty} xy e^{-x-y} \, dx \, dy \\
 &= \int_0^{\infty} x e^{-x} \cdot \int_0^{\infty} y e^{-y} \, dy = 1 \cdot 1 = 1
 \end{aligned}$$

$$E[X \cdot Y] = E[X] \cdot E[Y]$$

$\Rightarrow$ X and Y are independent random variable.

$$\text{Cov}[X, Y] = E[X \cdot Y] - E[X] \cdot E[Y] = 0$$

$$\text{Correlation} = \frac{\text{Cov}[X, Y]}{\sigma_X \sigma_Y} = 0$$

4(b) Given $n_1 = 16$ $\mu_1 = 75$ $\sigma_1 = 8$
 $n_2 = 9$ $\mu_2 = 70$ $\sigma_2 = 12$

$$P(\bar{x}_1 - \bar{x}_2 > 4) = P\left(\frac{\bar{x}_1 - \bar{x}_2 - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} > \frac{4 - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} \right)$$

$$P[\cancel{Z > 4}] = P\left[Z > \frac{4 - (75 - 70)}{\sqrt{\frac{64}{16} + \frac{144}{9}}} \right]$$

$$= P\left[Z > \frac{-1}{\sqrt{4 + 16}} \right]$$

$$= P[Z > -0.2236]$$

$$= 0.5871. \quad \underline{\underline{\text{Ans.}}}$$

6.10) Given $n_1 = 8$, $n_2 = 10$, $\sum (x - \bar{x})^2 = \frac{84.4}{7}$
 $\sum (y - \bar{y})^2 = 102.6$
 $\therefore S_{x^2} = \frac{1}{n_1 - 1} \sum (x - \bar{x})^2 = \frac{84.4}{7} = 12.057$
 $S_{y^2} = \frac{1}{n_2 - 1} \sum (y - \bar{y})^2 = \frac{102.6}{9} = 11.4$

Suppose that the two samples are from population ~~from~~ with the same variance. Then

$$F = \frac{S_{x^2}}{S_{y^2}} = \frac{12.057}{11.4} = 1.057$$

Tabulated value $F_{0.05}(7,9) = 3.5329$

Since $1.057 < 3.29$, the difference is not significant at 5% level.

Ques. 6. (b)

Observation are: 6, 5, 1, 2, 2, 5, 7, 5, 3, 7, 4, 7.

$$H_0: \text{Median} = 3 = M, \quad H_1: \text{Median} \neq 3$$

$$d_i = x_i - M_e$$

$$= 3, 2, -2, -1, -1, 2, 4, 2, 0, 4, 1, 4$$

write in ascending order [in magnitude value]

$$= \begin{matrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 & 11 \\ -2, & -1, & -1, & 1, & 2, & 2, & 2, & 3, & 4, & 4, & 4 \end{matrix}$$

$$\text{Rank}, \quad -1, -2.5, -2.5, 4, 6, 6, 6, 8, 10, 10, 10$$

$$T^+ = 4, 6$$

Ascending order

$$\begin{matrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 & 11 \\ 1, & 1, & 1 & 2 & 2, & 2, & 2 & 3, & 4, & 4, & 4, \end{matrix} \quad n=11$$

~~Rank~~

$$-1, -1, +1, -4.5, +4.5, +4.5, +4.5, +8, +10, +10, +10$$

$$T^+ = 1, 4.5, 4.5, 4.5, 8, 10, 10, 10 = 48$$

$$T^- = -1, -1, -4.5$$

$$= 6$$

(absolute sum)

$$T = \min \{T^+, T^-\}$$

$$= \min \{48, 6\} = 6$$

$T < T_\alpha \Rightarrow H_0$ is rejected ie the test is not s.
Median number of hours of student studies before a

Ques. 7(a) Taking the hypothesis that the experimental result support the theory, the expected frequencies are respectively

$$\frac{9}{16} \times 1600, \frac{3}{16} \times 1600, \frac{3}{16} \times 1600, \frac{1}{16} \times 1600$$

ie 900, 300, 300, 100. Thus.

Group	O_i (Observed Frequency)	E_i (Expected Frequency)	$(O_i - E_i)^2$	$\frac{(O_i - E_i)^2}{E_i}$
I	882	900	$(882 - 900)^2$	$\frac{(882 - 900)^2}{900}$
II	313	300	$(313 - 300)^2$	$\frac{(313 - 300)^2}{300}$
III	287	300	$(287 - 300)^2$	$\frac{(287 - 300)^2}{300}$
IV	118	100	$(118 - 100)^2$	$\frac{(118 - 100)^2}{100}$

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$$

$$= 0.3600 + 0.5633 + 0.5633 + 3.24$$

$$= 4.7266$$

Degree of freedom = $4 - 1 = 3$, $\chi^2_{0.05, 3} = 7.815$
5% of level.

Since calculated value < the tabular value.
Accept H₀. Experimental values support the theory.

Ques. 7. (b) This is the case of two samples are drawn from the same population.

Given: $n_1 = 1000$, $n_2 = 2000$

$$\bar{x}_1 = 67.5 \quad \bar{x}_2 = 68.2 \quad \sigma = 2.5$$

Null Hypothesis H_0 : Samples are drawn from the same population i.e. $\mu_1 = \mu_2$ and $\sigma_1^2 = \sigma_2^2$
 $= \sigma^2 = (2.5)^2$

Alternative Hypothesis: $\mu_1 \neq \mu_2$

Test statistic $Z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\sigma^2 \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}} \sim N(0,1)$

$$Z = \frac{67.5 - 68.0}{\sqrt{(2.5)^2 \left(\frac{1}{1000} + \frac{1}{2000} \right)}} = \frac{-0.5}{2.5 \times 0.0381} = -5.1$$

$$|Z| = 5.1$$

Conclusion $Z_{0.025} = 1.96$,

Since $|Z| > 1.96$, therefore it is highly significant and we reject the null hypothesis at all the level of significance and conclude that the samples are certainly drawn from same population.

8.a. Solve by Simplex method.

$$\text{Max } Z = 5x_1 + 3x_2, \text{ Subject to } \begin{aligned} x_1 + x_2 &\leq 2 \\ 5x_1 + 2x_2 &\leq 10 \\ 3x_1 + 8x_2 &\leq 12 \\ x_1, x_2 &\geq 0 \end{aligned}$$

Standard form $\text{Max } Z = 5x_1 + 3x_2 + 0s_1 + 0s_2 + 0s_3$

Subject to $x_1 + x_2 + s_1 + 0s_2 + 0s_3 = 2$

$$5x_1 + 2x_2 + 0s_1 + s_2 + 0s_3 = 10$$

$$3x_1 + 8x_2 + 0s_1 + 0s_2 + s_3 = 12$$

$$x_1, x_2, s_1, s_2, s_3 \geq 0.$$

$$\begin{aligned} s_1 &= 2 \\ s_2 &= 10 \\ s_3 &= 12 \end{aligned} \left| \begin{array}{l} \text{Initial} \\ \text{B.S.} \end{array} \right.$$

First Simplex Table.

		C_j	5	3	0	0	0	min Ratio
B	C_B	x_B	x_1 x_4	x_2 x_5	s_1 y_1	s_2 y_2	s_3 y_3	
s_1	0	2	1	1	1	0	0	2 ($i=1$)
s_2	0	10	5	2	0	1	0	2 ($i=2$)
s_3	0	12	3	8	0	0	1	4
$\sum C_B x_i$			0	0	0	0	0	
$\Delta_j = C_j - \sum C_B x_i$			5	3	0	0	0	

Thus, the problem is the problem of degeneracy.

We will proceed as follows.

1. Renumber the columns of the table such that the columns associated to the identity matrix comes first in the proper order.

$$S_1 = \gamma_1, \quad S_2 = \gamma_2, \quad S_3 = \gamma_3, \quad x_1 = \gamma_4, \quad x_2 = \gamma_5$$

2. Since the minimum occurs at $i=1, i=2$

$$\therefore I_1 = \{1, 2\}.$$

and incoming vector is $x_1 = \gamma_4, \therefore t=4$

For $i=1, 2$

$$\min_{i \in I_1} \left\{ \frac{y_{i1}}{y_{it}}, y_{it} > 0 \right\} = \min \left\{ \frac{y_{11}}{y_{14}}, \frac{y_{21}}{y_{24}} \right\}$$

$$= \min \left(\frac{1}{1}, \frac{0}{5} \right) = \min \{1, 0\} = 0 = \frac{y_{21}}{y_{24}}$$

This minimum is unique and occurs $i=2$

$\therefore$ The vector to be deleted from the basis is $\gamma_2 = S_2$

B	C_B	x_B	x_1	x_2	S_1	S_2	S_3	Min
S_1	0	6	0	$3/5$	1	$-1/5$	0	$0 \rightarrow$
x_1	5	2	1	$2/5$	0	$1/5$	0	5
S_3	0	6	0	$34/5$	0	$-3/5$	1	$15/17$
$Z_j = \sum C_B x_B = 10$			5	2	0	1	0	
$\Delta_j = C_j - Z_j$			0	1	0	-1	0	

			q_j	5	3	0	0	0	
B	C_B	x_B	x_1	x_2	s_1	s_2	s_3	Min ratio	
x_2	3	0	0	1	$5/3$	$-1/3$	0		
x_1	5	2	1	0	$-2/3$	$1/3$	0		
s_3	0	6	0	0	$-3\frac{1}{3}$	$5/3$	1		
$Z_j = \sum C_B x_B = 10$			5	3	$5/3$	$2/3$	0		
$\Delta_j = q_j - Z_j$			0	0	$-5/3$	$-2/3$	0		

Since all $\Delta_j \leq 0$, stop the process.

$$x_1 = 5, \quad x_2 = 3, \quad \max Z = \underline{10}$$

8. (b) Write the dual of the problem.

$$\text{Min } W = 2y_1 + 4y_2 + 3y_3$$

$$\text{Subject to } -y_1 + y_2 + y_3 \geq 2$$

$$2y_1 + y_2 + 0y_3 \geq 1$$

$$\text{Restriction } y_1, y_2, y_3 \geq 0$$

Write the problem in standard form (Matrix form).

$$\text{Min } W = (2, 4, 3) [y_1, y_2, y_3]$$

Subject to $AX \geq B$

$$\begin{bmatrix} -1 & 1 & 1 \\ 2 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \geq \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

Dual of the problem.

$$\text{Max } Z = B'X = (2, 1) (x_1, x_2) \\ = 2x_1 + x_2$$

Subject to $A'X \leq C'$

$$\begin{bmatrix} -1 & 2 \\ 1 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \leq \begin{bmatrix} 2 \\ 4 \\ 3 \end{bmatrix}$$

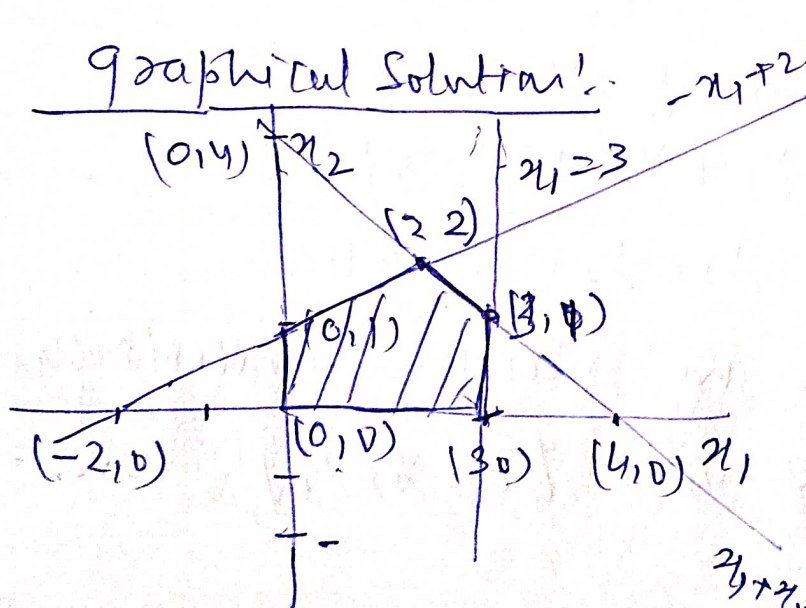
Dual Problem is $\text{Max } Z = 2x_1 + x_2$

Subject to $-x_1 + 2x_2 \leq 2$

$$x_1 + x_2 \leq 4$$

$$x_1 + 0x_2 \leq 3; \quad x_1, x_2 \geq 0$$

Graphical Solution:



Points are

Points	$Z = 2x_1 + x_2$
$(0,0)$	0
$(3,0)$	6
$(0,1)$	1
$(3,1)$	7
$(2,2)$	6

$\text{Max } Z = 7, \quad x_1 = 3, \quad x_2 = 1$

9. (a) Transportation problem

	P_1	P_2	P_3	P_4	Requirement
M_1	19	14	23	11	11
M_2	15	16	12	21	343
M_3	30	25	16	39	197
Availability	6	10	12	15	3

Availability

4	2	4	10
15	9	4	18
15	9	4	
	9	4	

Solution is.

	P_1	P_2	P_3	P_4
M_1	19	14	23	11
M_2	15	16	12	21
M_3	30	25	16	39

$$\begin{aligned}
 \text{Transportation Cost} &= 11 \times 11 + 15 \times 6 + 21 \times 4 + 25 \times 7 + 16 \times 12 \\
 &= 121 + 90 + 84 + 175 + 192 \\
 &= 662
 \end{aligned}$$

Optimality condition

$$m+n-1 = \text{No. of allocation}$$

$$3+4-1 = 6 = 6 \quad \text{Verified!}$$

Modi method! · U-U method ·

19) 15 +	14) 6 +	23) 3 +	11) 11 ⑪	$u_1 = -10$
15) 6 ⑥	14) 3 ③	12) 7 +	21) 9 ⑨	$u_2 = 0$
30) 24 +	25) 7 ⑦	16) 12 ⑫	39) 30 +1	$u_3 = 9$

$$v_1 = 15 \quad v_2 = 16 \quad v_3 = 7 \quad v_4 = 21$$

Here all $\Delta_j \geq 0$ +ive. Hence VAM is also optimum solution.

Optimum cost is = 662

Q. (b)

	U	V	W
A	17	25	31
B	16	25	16
C	12	14	11

Step 1. row minima. Step 2 column minima.

0	8	14
0	15	6
1	3	0

0	5	11
0	12	3
1	0	0

Step 3. Search the assignment-

0	5	11
0	12	3
1	0	0

Step 4.

0	2	8
0	9	0
4	0	0

Therefore $A \rightarrow U, B \rightarrow W, C \rightarrow V$

$$\text{Minimum cost} = 17 + 16 + 14 = 47$$

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Q 5 (a) mean $\bar{x} = 5 = \mu_x = 5$ by (C.I.T)

$$\sigma^2 = \frac{(6-4)^2}{12} = \frac{1}{3}$$

$$\sigma_{\bar{x}}^2 = \frac{\sigma^2}{n} = \frac{1}{3 \times 40} = \frac{1}{120} \text{ by (C.I.T)}$$

Q 3 (a) $n = 120$, $X_1 = 110$, $\mu = np = 120 \times 0.85 = 102 = \mu$

(i) ~~probability~~ $P(X > 110)$ $\sigma = \sqrt{npq} = 3.91$

$$= P\left(Z > \frac{(X_1 - \frac{1}{2}) - \mu}{\sigma}\right) \text{ (by correction)}$$

$$= P\left(Z > \frac{(110 - \frac{1}{2}) - 102}{3.91}\right) = P\left(Z > \frac{109.5 - 102}{3.91}\right)$$

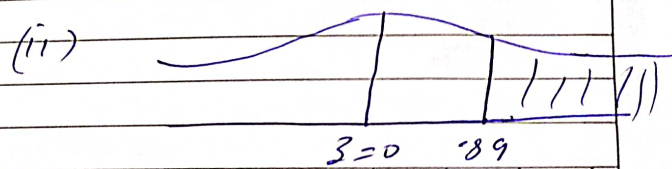
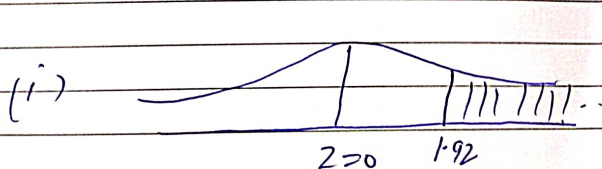
$$= P(Z > 1.92) = 0.0274$$

(ii) $X = 105$, $\sigma = 3.91$, $np = \mu = 102$

$$P(X < 105) = P\left(Z < \frac{(105 + \frac{1}{2}) - 102}{3.91}\right) \text{ (correct)}$$

$$\therefore P(10 < 3 \text{ etc}) = P(a < 3 \text{ etc})$$

$$P\left[Z < \frac{(105 + \frac{1}{2}) - 102}{3.91}\right] = P(Z < 0.89) = 0.1867$$



$$P(Z > 1.92)$$

$$= 0.5 - 0.4726 = 0.0274$$

$$P(Z < 0.89)$$

$$= 0.5 + 0.3133$$

$$= 0.8133$$

Teacher's Signature :

$$95(b) \quad n = 6 \quad (6 \text{ balls})$$

$$p_1 = 5/12 = 0.416 \quad (\text{prob. of choosing red})$$

$$p_2 = 4/12 = 0.333 \quad (\text{prob. of choosing a white})$$

$$p_3 = 3/12 = 0.25 \quad (\text{prob. of choosing a blue})$$

$$n_1 = 3 \quad (3 \text{ red marbles chosen})$$

$$n_2 = 1 \quad (1 \text{ white marble chosen})$$

$$n_3 = 2 \quad (2 \text{ blue marbles chosen})$$

$$k = 3 \quad (3 \text{ possibilities})$$

$$p = \frac{L^n}{L^{n_1} \times L^{n_2} \times L^{n_3}} \times (p_1^{n_1} \times p_2^{n_2} \times p_3^{n_3})$$

$$p = \frac{26}{23 \times 21 \times 22} (.416^3 \times .333^1 \times .25^2)$$

$$p = 60 \times 0.0720 \times 0.333 \times 0.0625$$

$$p = 0.0899$$